Himanshu Chaudhary

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Education

California Institute of Technology 2026
Ph.D. (Physics) *Pasadena, US*
Advisor: Saul Teukolsky and Mark Scheel
Thesis: "High-Accuracy Binary Black Hole Simulations in SpEC"

Indian Institute of Science 2020
MS in Physics *Bengaluru, India*
Advisor: Chethan Krishnan

Indian Institute of Science 2019
BSc (Physics major, Mathematics minor) *Bengaluru, India*
Advisor: Rahul Pandit

Publications

1. **Himanshu Chaudhary**, Mark A. Scheel, Robert Owen, Saul A. Teukolsky, Michael Boyle, Lawrence E. Kidder, and Harald P. Pfeiffer, "How accurate is Spectral Einstein Code (SpEC)? And what limits its accuracy?" (arXiv).

2. **Himanshu Chaudhary**, Rob Owen, Mark A. Scheel, and Saul A. Teukolsky, "Finding black hole spins efficiently during a numerical binary evolution," arXiv:2608.12211.

3. **Himanshu Chaudhary** and Chethan Krishnan, "Cosmic Censorship of Trans-Planckian Field Ranges in Gravitational Collapse," SciPost Phys. 9, 036 (2020).

4. Daniel G. Cocks, **Himanshu Chaudhary**, and Joshua R. Machacek, "Benchmark for the Determination of the Positronium Formation Fraction in Interstellar Media," ApJ 878, 123 (2019).

5. **Himanshu Chaudhary**, Biplab Mahato, Lakshya Priyadarshi, Naman Roshan, Utkarsh, and Apoorva D. Patel, "A Software Simulator for Noisy Quantum Circuits," arXiv:1908.05154.

Fellowships and Awards

– KVPY Fellowship 2015–2020

– Chennupati and Vidya Jagadish Fellowship and Scholarship Awards 2019

Conference Presentations

8. **NR Community Call** September 2026
   LISA-accurate simulations in SpEC

7. **NR Community Call** September 2025
   How accurate is SpEC?

6. **PCGM 41** April 2025
   How accurate is SpEC?

5. **APS Global Physics Summit** March 2025
   Testing the accuracy limit of binary black hole simulations in SpEC

4. **APS April Meeting** April 2024
   Efficient high-mass-ratio binary-black-hole simulations using new gauges and initial data

3. **PCGM 40** March 2024
   Efficient high-mass-ratio binary-black-hole simulations using new gauges and initial data

2. **APS April Meeting** April 2023
   Gauge conditions for Generalized Harmonic Evolution that reduce the deformations of the apparent horizon

1. **PCGM 39** April 2023
   Gauge conditions for Generalized Harmonic Evolution that reduce the deformations of the apparent horizon

Teaching and Mentoring

– **Teaching Assistant**, California Institute of Technology 2021
  Ph 1b: Classical Mechanics and Electromagnetism (Practical Track)
  Led recitation sections and prepared and reviewed exam questions.
– **Student Mentoring** — *What Does a Black Hole Look Like?*
  Mentored Arhanth Kurli in using relativistic ray tracing to investigate light propagation near black holes.

Software Contributions

– **SpEC** — Developed faster spin-computation algorithms and high-accuracy workflows for binary-black-hole simulations.
– **Qiskit-Aakash** — Contributed to a quantum-computing simulator for noisy quantum circuits using IBM Qiskit as its backend.

Programming Proficiency

Primary Python, C++
Secondary Julia, Mathematica, MATLAB